

AMAZING SPACE FACTS

Displays in the sky called the Aurora Borealis and Aurora Australis are caused when electrically charged particles from the Sun are trapped by Earth's magnetic poles and collide with gas atoms in the atmosphere over the planet's polar regions. The displays create shifting, coloured light displays in the sky.

We see a total solar eclipse when the Moon precisely covers up the Sun. By an amazing coincidence, the Sun is 400 times bigger than the Moon, and also 400 times further away.

Shooting stars are not stars at all. They are tiny fragments of space dust that burn up in Earth's atmosphere, leaving a trail of light behind them.

If you see a slow-moving dot in the sky, it may be a satellite, lit up by the Sun. You may even spot the International Space Station (ISS).

The furthest object visible to the naked eye is the Triangulum Galaxy, or M33, three million light years away – although you need to be a long way away from lights to be able to see it.

Imagine that you were in a car, travelling at 110 kph (68 mph). To reach the Sun would take 155 years; the nearest star, Proxima Centauri, 2 million years; and the centre of our Milky Way galaxy, 14 billion years.

Space starts only 100 km (62 miles) above our heads – a one-hour drive if your car could go straight upwards.

Pieces of "space junk" – including broken up satellites and tools dropped by astronauts – travel 10 times faster than a bullet, and are very dangerous to spacecraft in space. In 2009, space junk destroyed a communications satellite.

When the Sun dies, five billion years from now, humankind will have to find a new planet circling another star. No problem, there are over 3,000 of them out there.

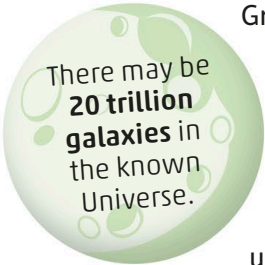
A British backyard astronomer, Tom Boles, has discovered a record 155 supernovae.

The world's biggest telescope, the Gran Telescopio Canarias in the Canary Islands, Spain, collects light with a mirror half the size of a tennis court.

Much of the matter in the Universe, including that making up our bodies, is recycled stardust.

Five Solar System planets are easily visible to the naked eye. They are among the brightest objects in the night sky.

Halley's Comet, which visits the Sun every 76 years, was the first comet to be predicted by Newton's law of gravity. Edmond Halley first noticed this, which is why the comet is named after him.



There may be
20 trillion
galaxies in
the known
Universe.